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VANTAGE

Portfolio-aware financial risk intelligence

From financial news to explainable portfolio stress signals

Candidate: Rhythm Bhetwal

College / Campus: NIT Meghalaya

Prototype for educational use - synthetic examples only

News is not portfolio risk

The same headline can have very different implications for different holdings.

THE GAP

Generic sentiment ignores portfolio composition.

Free-form AI can produce unverifiable loss numbers.

Static alerts are hard to audit or prioritize.

OUR APPROACH

Classify event and sentiment using local ONNX models.

Match event to portfolio holdings and sectors.

Calculate stress and loss with deterministic formulas.

WHAT CHANGES

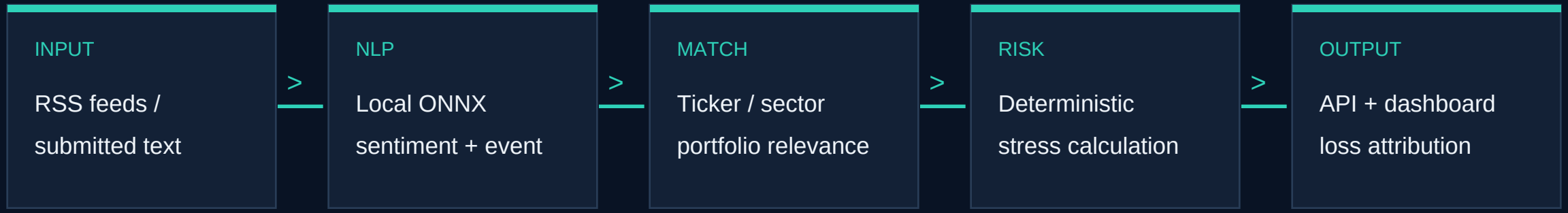
A traceable chain from headline to exposure.

Relevant events are ranked for the portfolio.

Loss attribution is visible by asset class.

One explainable data flow

AI classifies text; deterministic services calculate numerical risk.



PostgreSQL persists portfolios, events, and stress-test runs.

Optional semantic enrichment is separate from risk calculations.

Architecture: docs/assets/architecture_flow.svg

Designed for local, auditable analysis

Core numerical outputs do not depend on a generative AI response.

Local NLP

FinBERT sentiment + 8-class event classifier

ONNX inference via transformers.js

Quantitative engine

Pure TypeScript deterministic risk engine

Inspectable stress and loss calculations

Application stack

React + TypeScript + Vite frontend

Node.js + Express + Prisma backend

Persistence and enrichment

PostgreSQL for prototype state

Gemini semantic enrichment is optional

Measured model checks; no claimed business uplift

Reported internal validation figures documented with the model artifacts.

2,375 / 2,375	0	< 0.000050	0.9281
matching ONNX predictions	label mismatches	maximum probability difference	event-classifier macro F1

Prototype output: event category, sentiment, portfolio relevance, stress parameters, and asset-class loss attribution.

Validation is not proof of live-market accuracy or investment performance.

Turn alerts into an explainable risk workflow

Designed to support analyst review, not replace it.

PRIORITIZE

Focus attention on events that match holdings, sectors, or portfolio-wide exposures.

TRACE

Show how event category, portfolio exposure, and stress parameters connect.

COMPARE

Inspect estimated losses by asset class to support scenario discussion and review.

Business impact is qualitative; no time or financial savings are asserted.

Assumptions, risks, and next steps

Use only synthetic/public example data; review all outputs independently.

LIMITATIONS

Stress formulas are simplified estimates.
Entity matching can miss context or corporate relationships.
Live ingestion depends on public feeds and network availability.
This is not investment advice.

NEXT STEPS

Add historical scenario backtesting.
Expand entity resolution and coverage.
Evaluate with broader, sourced public data.
Record a local end-to-end demo and complete candidate submission details.